

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A warehouse stores the quantity of products available in eight storage racks. Before dispatch, the warehouse manager wants the processor to arrange the quantities in ascending order and identify the rack with the lowest stock for replenishment. Write an 8085 ALP to
 - Sort the quantities in ascending order.
 - Store the sorted values.
 - Identify and store the minimum quantity.
2. During data transmission, a block of eight bytes is copied from one memory region to another. To ensure successful transmission, each byte in the destination memory must be compared with its corresponding source byte. Write an 8085 ALP to
 - Copy the data block.
 - Compare each source byte with the copied byte.
 - Count the number of mismatches.
 - Store the mismatch count.
3. A communication controller receives an 8-bit binary code. According to the communication protocol, the code is valid only if it contains an even number of 1s. Develop an 8085 ALP to
 - Count the number of set bits.
 - Determine whether the parity is even or odd.
 - Store a validity flag in memory.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

1.

Load me at

1

LXI H,8050

2

MVI B,07

3

4

LOOP2: MOV C,B

5

LXI H,8050

6

7

LOOP1: MOV A,M

8

INX H

9

CMP M

10

JC NEXT

11

JZ NEXT

12

13

MOV D,M

14

MOV M,A

15

DCX H

16

MOV M,D

17

INX H

18

19

NEXT: DCR C

20

JNZ LOOP1

21

22

DCR B

23

JNZ LOOP2

24

25

LXI H,8050

26

MOV A,M

27

STA 8060

28

29

HLT

Data

Stack

Keypad

Memory

Start

8050

Address (Hex)	Address	Data
1F72	8050	9
1F73	8051	12
1F74	8052	20
1F75	8053	20
1F76	8054	45
1F77	8055	55
1F78	8056	67
1F79	8057	81
1F7A	8058	0
1F7B	8059	0
1F7C	8060	9
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0
1F80	8064	0

Line No	Assembler Message
0	Program assembled successfully

2.

Load me at

1 LXI H,8050

2 LXI D,8060

3 MVI B,08

4 MVI C,00

5

6 LOOP: MOV A,M

7 STAX D

8

9 LDAX D

10 CMP M

11 JZ NEXT

12

13 INR C

14

15 NEXT: INX H

16 INX D

17 DCR B

18 JNZ LOOP

19

20 MOV A,C

21 STA 8070

22

23 HLT

Data Stack KeyPad Memory I/O

Start 8050

Address (Hex)	Address	Data
1F72	8050	11
1F73	8051	12
1F74	8052	33
1F75	8053	44
1F76	8054	55
1F77	8055	66
1F78	8056	77
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	11
1F7D	8061	12
1F7E	8062	33
1F7F	8063	44

Line No	Assembler Message
0	Program assembled successfully

1F7A	8058	0
1F7B	8059	0
1F7C	8060	11
1F7D	8061	12
1F7E	8062	33
1F7F	8063	44
1F80	8064	55
1F81	8065	66
1F82	8066	77
1F83	8067	0
1F84	8068	0
1F85	8069	0
1F86	8070	0

3.

Load me at

1 LDA 8050

2 MVI B,08

3 MVI C,00

4

5 LOOP: RRC

6 JNC NEXT

7

8 INR C

9

10 NEXT: DCR B

11 JNZ LOOP

12

13 MOV A,C

14 ANI 01

15 JZ EVEN

16

17 MVI A,01

18 STA 8060

19 HLT

20

21 EVEN: MVI A,00

22 STA 8060

23

24 HLT

Data

Stack

KeyPad

Memory

I/O Ports

Start

OK

Address (Hex)	Address	Data
1F72	8050	55
1F73	8051	0
1F74	8052	0
1F75	8053	0
1F76	8054	0
1F77	8055	0
1F78	8056	0
1F79	8057	0
1F7A	8058	0
1F7B	8059	0
1F7C	8060	1
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No	Assembler Message
0	Program assembled successfully